Ibrahim Hassan

Fullstack Developer (.NET, Angular)

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Professional Summary

Fullstack Developer (.NET, Angular) and ECPC Finalist with hands-on experience building real-time platforms, encrypted authentication systems, and multi-service architectures using ASP.NET Core, Clean Architecture, and Angular.

Technical Projects

Wasla – Smart Microbus Platform (GitHub · Live Demo) Jan 2026 – Jun 2026

- **11 controllers, 16 domain entities, and 86 DTOs** across 3 architectural layers: built a real-time transportation management platform using **ASP.NET Core** and **Clean Architecture**.
- **4 concurrent SignalR hubs** engineered to push live queue positions, driver status changes, GPS coordinates, and route-level ETAs to mobile and web clients simultaneously.
- **4 real-time broadcasts triggered per scan**: implemented AES-GCM encrypted QR authentication where a single scan decrypts, validates, mutates the queue, and creates a trip atomically.
- Designed a GPS pipeline using **OSRM routing**, **Haversine filtering**, and **Redis caching** (4h TTL); deployed with **Docker Compose** (API + SQL Server + Redis).

WhatsApp Messaging Infrastructure (Gateway · Engine · Dashboard) 2026

- Built an **ASP.NET Core orchestration gateway** bridging the main backend to a headless WhatsApp engine, with message delivery logging and server-to-server auth via custom action filters.
- Developed a self-healing **headless WhatsApp engine** with a 6-state state machine, heartbeat-based zombie detection, and exponential backoff recovery for production resilience.
- Created a real-time **Angular monitoring dashboard** using Standalone Components and Signals, integrating SignalR for live QR display, with JWT auth flow, route guards, and auto-reconnect logic.

E-StoreX – Advanced E-Commerce Frontend (GitHub · Live Demo) Dec 2025 – Feb 2026

- **20+ modular features** architected in an Angular frontend using Standalone Components and clean layered architecture.
- **40% reduction** in change detection cycles achieved through Signal-based state management.
- **4-step Stripe checkout** flow implemented, handling payment intents and real-time order creation.
- **50% reduction** in First Contentful Paint (FCP) achieved via SSR and Lazy Loading; built a responsive TailwindCSS UI with dark/light themes and cart persistence.

E-StoreX – Enterprise E-Commerce API (GitHub · Live Demo) 2025

- **140+ endpoints** developed in a Clean Architecture Web API covering products, orders, payments, and user management.
- Real-time payment processing integrated via **Stripe API with Webhooks** for secure, automated order confirmation.
- **4+ external services orchestrated**, including Stripe, Redis caching, Hangfire background jobs, and email notifications.
- **3 authentication providers** (JWT, Google OAuth, GitHub OAuth) built into a security layer with custom rate limiting middleware.
- **250+ automated xUnit test cases** achieving 90%+ code reliability across unit and integration testing.

Professional Experience

Full Stack .NET Developer Oct 2025 – Nov 2025

Laundry Management System – Freelance *Cairo, Egypt*

- Designed and developed a complete **ASP.NET Core MVC** application, delivering both backend services and responsive user interfaces for a laundry management system.
- Implemented **role-based access control** using ASP.NET Identity with Admin, Owner, and Employee roles, en-

suring secure authentication and authorization.
- Developed business modules for invoices, expenses, orders, customers, and financial management with **Entity Framework Core** and SQL Server.
- Built dynamic **Arabic/English MVC views**, automated PDF/Excel reporting, and integrated WhatsApp messaging to streamline business operations.

Technical Skills

Programming
C#, Java, C++, JavaScript, TypeScript, SQL

Backend
ASP.NET Core, Web API, Minimal APIs, Entity Framework Core, MVC, LINQ, Hangfire, MediatR (CQRS), SignalR, Node.js

Frontend
Angular, Signals, RxJS, TypeScript, TailwindCSS, SSR, HTML5, CSS3, SCSS

Databases
SQL Server, Redis, PostgreSQL

Auth & Security
JWT, OAuth 2.0, ASP.NET Identity, AES-GCM Encryption, Role-Based Access Control

Testing
xUnit, Moq, FluentAssertions, Integration Testing

Tools & Practices
Git, GitHub, Docker, Docker Compose, Azure, Swagger, Serilog, AutoMapper, Clean Architecture, REST APIs

AI Tooling
Claude, GitHub Copilot, ChatGPT

Soft Skills
Problem-Solving, Cross-Functional Team Collaboration, Technical Communication, Adaptability

Achievements & Certifications

- **ECPC Finalist** – Egyptian Collegiate Programming Contest (2024)
- **AAIB Banking Internship** – Arab African International Bank (2025)
- **C# Development** – Udemy Certificate (2025)
- **ASP.NET Core Web API** – Udemy Certificate (2025)
- **Angular Development** – Udemy Certificate (2025)

Education

Bachelor of Information Technology — Sep 2022 – Jun 2026
Faculty of Science, Minia University — *Minia, Egypt*

Languages

Arabic *(Native)* | English